

Ahmad Rahman

Senior AI Engineer | Health AI | LLMs | Full-Stack | Cloud

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Senior AI engineer with 6+ years building production LLM systems end-to-end, from RAG architectures and structured output pipelines to evaluation, deployment, and governance. Shipped a domain-constrained conversational AI achieving 3.9x faster task completion than traditional retrieval in published evaluation; an AI clinical documentation system cutting paramedic report time 4-6x; and an LLM education platform deployed to 264 learners across two universities. Published researcher with papers at Mayo Clinic Proceedings: Digital Health and NEJM Catalyst.

EXPERIENCE

Virtual Health Hub, University of Saskatchewan Software Technology Lead

Feb 2024 - Present

Led architecture, development, and deployment of AI-powered and data-driven software systems for rural and remote virtual care across Saskatchewan.

- Built Echo, a production RAG-based conversational AI assistant serving rural healthcare planners across Saskatchewan. Designed and implemented record-aware chunking over structured VHH datasets, embedding generation with Vertex AI text-embedding-005, vector storage and retrieval via Firestore native vector search, and generation with Gemini 3 Flash.
- Engineered multi-layer domain-constraint architecture for Echo combining retrieval-boundary enforcement, application-side context construction, prompt-level refusal patterns, and provenance surfacing, enabling safe deployment in a regulated health context.
- Designed and ran a scenario-based evaluation study of Echo against traditional information retrieval workflows: median task completion time dropped from 31s to 8s (3.9x speedup) across paired observations; mean accuracy score of 0.77 with 92.8% of responses fully or partially correct (n=98).
- Built EchoPCR, an AI clinical documentation system for paramedics using Deepgram streaming speech-to-text and LLM-driven structured JSON generation for patient care reports. Pre-trial results: documentation time reduced from 45-120 minutes to 10-20 minutes per report (4-6x). Now in active trial with 10 paramedics in La Ronge EMS.
- Architected and delivered VHH PATH, a 10-module digital health ecosystem (Vue/Quasar, Firebase, GCP) used by practitioners, planners, and administrators province-wide. Modules include community profiles (SCOPE), facility readiness (FRAME), technology repository (TRACE), MCDA-based decision support (SELECT), and AI assistant (Echo).
- Led VHH cloud and AI strategy across GCP (Vertex AI, Cloud Functions, Firestore), Azure, and AWS, including sovereign AI evaluation, enterprise LLM procurement, responsible AI governance, and data residency planning.
- Member of the eHealth Saskatchewan EMR interoperability task force; contributed to HL7/FHIR planning, ClinicianViewer integration, and TELUS Med Access SSO scope.
- Mentored and onboarded software developers; led cross-functional delivery across product, research, clinical, and leadership stakeholders.
- Co-authored 8 research publications across health AI, virtual care, and clinical documentation, including two published at Mayo Clinic Proceedings: Digital Health and one submitted to NEJM Catalyst Innovations in Care Delivery. Lead technical author on Echo and EchoPCR papers.

Core Syntax Co-Founder and CTO

Apr 2025 - Present

Co-founded an applied AI product studio building LLM-powered software across healthcare, education, and enterprise verticals.

- SPARK (spark.coresyntax.ai): Built an LLM-powered simulated patient platform for health professions education using GPT-5, Next.js, React, and Firebase. Deployed across two U15 Canadian universities with 264 learners to date. Designed instructor persona-authoring workflow, multi-turn conversational engine with persistent session state, cohort management, and learner interaction review.

- SREDER (sreder.ca): Built an AI-assisted SR&ED reporting platform for CRA-compliant technical documentation, reducing consultant report preparation time from 2-3 hours to under 20 minutes (7-9x). 50+ reports generated in production. Includes audio transcription pipeline, structured report generation aligned to CRA criteria, Stripe-based credit billing, and multi-tenant role-based access. 49% equity stake held by launch partner Advance-Tek.
- Manages partner equity model, go-to-market strategy, and full-stack technical delivery across both products.

EARLIER EXPERIENCE

University of Saskatchewan Programmer, Research Library and Data Centre Technology Aug 2021 - Jan 2024

Built AI/LLM pipelines and large-scale data infrastructure for major university research initiatives.

- Developed large-scale LLM and AI pipelines for the Archives Unleashed project; presented research at Internet Archives Conference (2023) and Access Conference (2022).
- Engineered high-performance data systems including databases, data warehouses, and scraping pipelines for major research initiatives; built and maintained ArcGIS and Solr-based search infrastructure.

Netpace Software Technical Lead

Apr 2019 - Apr 2020

- Led architecture and technology stack design for multiple enterprise products; reduced development timelines by approximately 20%.
- Built AI/ML models in Python analyzing 10TB+ of financial data, achieving approximately 25% improvement in predictive accuracy.

Netpace Software Engineer

Jun 2016 - Dec 2016

- Developed large-scale data processing systems using Hadoop and Hive for financial prediction; delivered big data training enabling adoption across engineering teams.

MSc, Computer Science University of Saskatchewan

May 2021

BSc, Computer Science NUCES, Karachi

2016 | Summa Cum Laude

RESEARCH AND PUBLICATIONS

- A Technology Selection Tool Applying Multiple-Criteria Decision Analysis for Virtual Care Implementation, Published, Mayo Clinic Proceedings: Digital Health. Technical contributor and co-author; built the MCDA-based decision-support software underlying the study.
- SCOPE and FRAME: Community and Facility Readiness Frameworks for Virtual Care Implementation, Published, Mayo Clinic Proceedings: Digital Health. Technical architect and co-author; built the underlying data systems.
- TRACE: Technology Repository for Virtual Care Evaluation, Submitted, NEJM Catalyst Innovations in Care Delivery. Co-author; designed and built the technology repository platform.
- Echo: A Domain-Constrained Conversational AI Assistant for Rural Health Planning, Submitted, Journal of Responsible Technology. Lead technical author; designed, built, and evaluated the system.
- SPARK: AI Simulated Patient Platform for Health Professions Education, In submission. Lead system architect and co-author. Live at spark.coresyntax.ai.
- FORCE: Force-Oriented Readiness and Change Estimator, Submitted. Co-author; contributed to a change management tool combining Force Field Analysis with Monte Carlo simulation to model virtual care implementation readiness under uncertainty.
- EchoPCR: AI-Assisted Clinical Documentation for Prehospital Care, In progress. Technical lead and co-investigator; designed and built the trial system.
- COST: Virtual Care Cost Modeling Study, In progress. Co-author; contributed cost-modeling logic and research design.

CORE TECHNOLOGIES

AI / ML: LLM APIs (OpenAI, Anthropic, Google Gemini, Cohere); RAG architecture and design; embedding generation and vector retrieval; structured output and function calling; streaming speech-to-text; multi-turn conversational AI; prompt engineering; domain-constrained AI; multi-layer guardrail design; evaluation methodology and statistical analysis; responsible AI governance

Cloud and Infrastructure: Google Cloud Platform (Vertex AI, Cloud Functions, Firestore, Firebase Auth and Hosting, BigQuery, Cloud Storage, Cloud Text-to-Speech); Microsoft Azure (Cosmos DB, App Service, hosting and identity); AWS (architecture, evaluation, and migration planning); serverless and event-driven systems; multi-tenant SaaS architecture; role-based access control; OAuth and JWT authentication; CI/CD workflows; Git and GitHub

Languages and Frameworks: Python, JavaScript, TypeScript, SQL; React, Next.js, Vue.js, Quasar, Tailwind CSS; Node.js, Express.js; REST API design; real-time and streaming systems; WebSockets

Healthcare and Compliance: HL7 and FHIR; EMR integration; PIPEDA-informed design; HIPAA-aware architecture; clinical data governance; encryption and audit logging; data residency planning

Engineering Practices: End-to-end product ownership; system architecture and technical leadership; cross-functional collaboration with clinical, research, and operations stakeholders; mentoring and hiring; agile delivery; technical writing and peer-reviewed publication